nature neuroscience

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https://doi.org/10.1038/s41593-020-0641-7

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Conservation of brain connectivity and wiring across the mammalian class

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Over 100 years ago, Ramon y Cajal hypothesized that two forces played a role in the evolution of mammalian brain connectivity: minimizing wiring costs and maximizing conductivity speed. Using diffusion MRI, we reconstructed the brain connectomes of 123 mammalian species. Network analysis revealed that both connectivity and the wiring cost are conserved across mammals. We describe a conservation principle that maintains the overall connectivity: species with fewer interhemispheric connections exhibit better intrahemispheric connectivity.

The mammalian brain is one of the most complex biological systems that have ever evolved. Most previous comparative studies of the mammalian brain concentrated on gray matter[1,2]. The connectome, the complete network of brain connectivity, is as important for brain functionality as the neurons, but only a few studies examined its evolution using an interspecies comparative approach[3–6]. We used diffusion magnetic resonance imaging (MRI) to reconstruct the macro-scale connectomes[7] of 123 mammalian species, from 12 orders, covering a range of brain volumes from ~0.1 to >1,000 ml (Fig. 1a and Supplementary Table 1). In all species, we normalized voxel size to brain volume, that is a similar number of voxels were acquired for all brains. We conducted several analyses to control for this decision (see Methods).

To validate our MRI-based method, we first replicated several previous results, including: (1) the power-law scaling of white matter as a function of brain volume[8] (Fig. 1b) and (2) the scaling of the commissural cross-sectional area with brain volume[9] (Supplementary Fig. 1). Such scaling laws mostly reflect geometric constraints such as the need both to connect distal brain regions in large brains and to pack the entire system in differently shaped skulls. We used network analysis to examine how brain size and functional demands interact in shaping mammalian brain connectivity[10–12]. We parcellated each brain into 200 areas ('nodes') across the cerebrum (including the cortical and subcortical regions; 100 nodes in each half; see Methods). We generated symmetric weighted networks, weighting each edge by the number of streamlines between the two nodes it connects, that is, we multiplied each edge by 1/(number of fibers between nodes), reflecting the probability of information transfer between the nodes.

It has been hypothesized that one of the main factors guiding the evolution of the connectome is a tradeoff between rapid information transfer and the cost of wiring[11,13,14]. On the one hand, more and longer connections are required for better connectivity, but, on the other, they increase the wiring costs. To examine this, we first quantified connectivity across mammals, using the network's mean-short path (MSP), a measure of global network communication[15]. The MSP reflects the minimal number of steps (connections) that need to be traversed between two nodes of the network. Note that the MSP is inversely related to connectivity, that is a high MSP reflects low connectivity. We found that the MSP weakly depends on brain volume, decreasing by only 40% from the largest brain to the smallest brain, even though brain volume decreased by more than four orders of magnitude (Fig. 1c and Supplementary Fig. 2). This weak dependency might be expected if the brain was a random graph, but the brain is very far from this (Supplementary Fig. 3).

How is global connectivity maintained despite the immense diversity in brain size and shape across mammals? The macro-scale mammalian connectome can be grossly divided into two hemispheric networks, which are connected to each other by commissural fibers and internally wired by intrahemispheric association fibers. The number and arrangement of commissural fibers vary greatly across mammals[9,16]. As we observed that the overall connectivity is constant across brains (Fig. 1c and Supplementary Fig. 2), we hypothesize that the intrahemispheric network in brains with fewer commissural connections must somehow be adapted to allow an overall similar connectivity. Note that here we examined the intrahemispheric MSP and not the global MSP, which was used for the previous analysis in Fig. 1. Importantly, because we always randomly sampled 4,000 hemispheric connections in all species (see Methods), and because all networks had the same number of nodes (200), any variation observed in intrahemispheric MSP should reflect actual changes in hemispheric connectivity.

Our results suggest a connectivity conservation principle, in which species with fewer (commissural) interhemispheric connections exhibit better intrahemispheric connectivity (Fig. 1d; there was a strong correlation between the proportion of commissural fibers and the intrahemispheric MSP, independent of brain volume; $P < 10^{-8}$, a generalized linear model (GLM) with the intrahemispheric MSP set as the explained parameter, and the commissural proportion and brain volume as fixed predictors). Results suggest that intrahemispheric connectivity compensates for poorer interhemispheric connectivity, maintaining the overall connectivity. Importantly, the proportion of commissural fibers and the intrahemispheric MSP are independent measurements, because the first was measured on the raw data, including all reconstructed tracts, whereas the latter was estimated on the hemispheric networks. Moreover, we achieved the same result when estimating the commissural proportion using its cross-sectional area, that is, without using the fiber-tracking algorithm (Supplementary Fig. 4). We also confirmed that this conservation phenomenon is independent of

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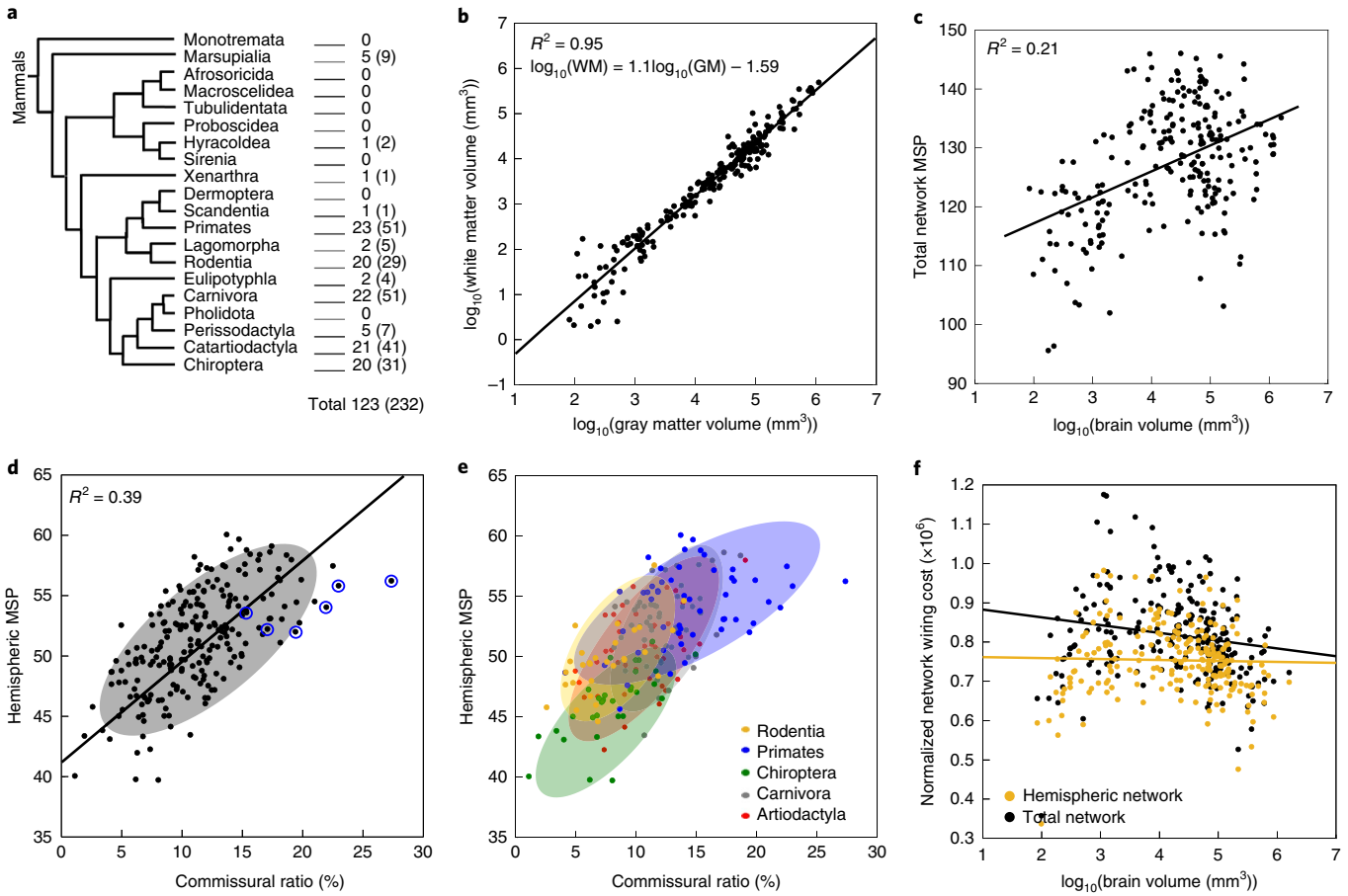


Fig. 1 | Global mammalian conservation of connectivity and wiring. a, The MaMI database includes scans of 123 species. Multiple individuals were scanned for some species; numbers shown in brackets. **b**, White matter (WM) volume versus gray matter volume on a log–log scale revealing a previously described linear correlation with a slope coefficient of ~1.19 ($P<10^{-10}$, Pearson's correlation *t*-test; $n=232$). **c**, Whole-brain MSPs as a function of total brain volume ($P<10^{-6}$, Pearson's correlation *t*-test, $n=232$). MSPs were calculated on weighted networks (weighted by 1/number of fibers) and thus the actual MSP values are on an arbitrary scale. Correlation *P* values are reported without a phylogenetic correction. **d**, Hemispheric MSP versus the proportion of commissural fibers for all mammals ($n=232$). The six human participants are labeled by blue circles. **e**, As in **d**, but with color coding of the five mammalian orders with more than 10 species. Ellipses in both **d** and **e** represent the 95% confidence intervals. **f**, The hemispheric (orange) and total wiring (black) cost (Pearson's correlation *t*-test, $R^2=0$, $P=0.58$ and $R^2=0.03$, $P=0.008$, for the hemispheric and total networks, respectively; $n=232$). The correlation of the total wiring cost is significant, but the explained variability is minimal. All statistical analyses in this figure were two tailed.

our analysis protocol and holds for both different scanning procedures and different network reconstruction and weighting parameters (Supplementary Figs. 5–11).

The mammalian brain includes five interhemispheric, commissural fiber systems, the main three of which are the corpus callosum, anterior commissure and hippocampal commissure. The division of interhemispheric connections between these three fiber systems is strongly species specific. The conservation principle that we found holds for all the mammalian orders that we examined, independent of their dominant commissural system (Fig. 1e). As we did not compensate for phylogeny in our correlation analyses, we ran the statistics for each order for which we had more than 10 species: Artiodactyla: $R=0.41$, $P=0.008$ ($n=40$); Carnivora: $R=0.33$, $P=0.019$ ($n=51$); Primates: $R=0.39$, $P=0.007$ ($n=52$); Chiroptera: $R=0.69$, $P=0.00001$ ($n=31$); Rodentia: $R=0.64$, $P=0.0001$ ($n=29$).

To examine the connectivity–wiring tradeoff, we quantified the total wiring length (that is, the sum of all fiber lengths). We used a normalized length, where each voxel has the same size relative to the brain volume across all mammals. Note that, as tractography does not actually measure axons but rather axonal bundles and fascicles, our length estimate reflects the wiring length of the macro-scale network. It is interesting that the normalized wiring length of all mammals did not depend on brain size (Fig. 1f).

Interhemispheric connections can also vary within a species. It is of interest that the connectivity conservation principle also holds within species. An examination of multiple individuals from four different species (humans, macaques, fruit bats and rats) revealed that individuals with fewer interhemispheric connections have better intrahemispheric connectivity (Fig. 2). The fact that our results also hold within species, where the scanning signal-to-noise ratio (SNR) and resolution were very similar across individuals, strongly suggests that the phenomenon we describe is not an artifact of the method.

To examine how intrahemispheric connectivity is improved in brains with fewer commissural fibers, we computed the proportion of strong connections in each brain, that is the proportion of edges with more than the median weight. Brains with better hemispheric connectivity had significantly more strong connections, suggesting that the stronger connections are major contributors to connectivity (Fig. 3a and Supplementary Fig. 11). Scrutiny of the network of strong connections revealed that the posterior–anterior

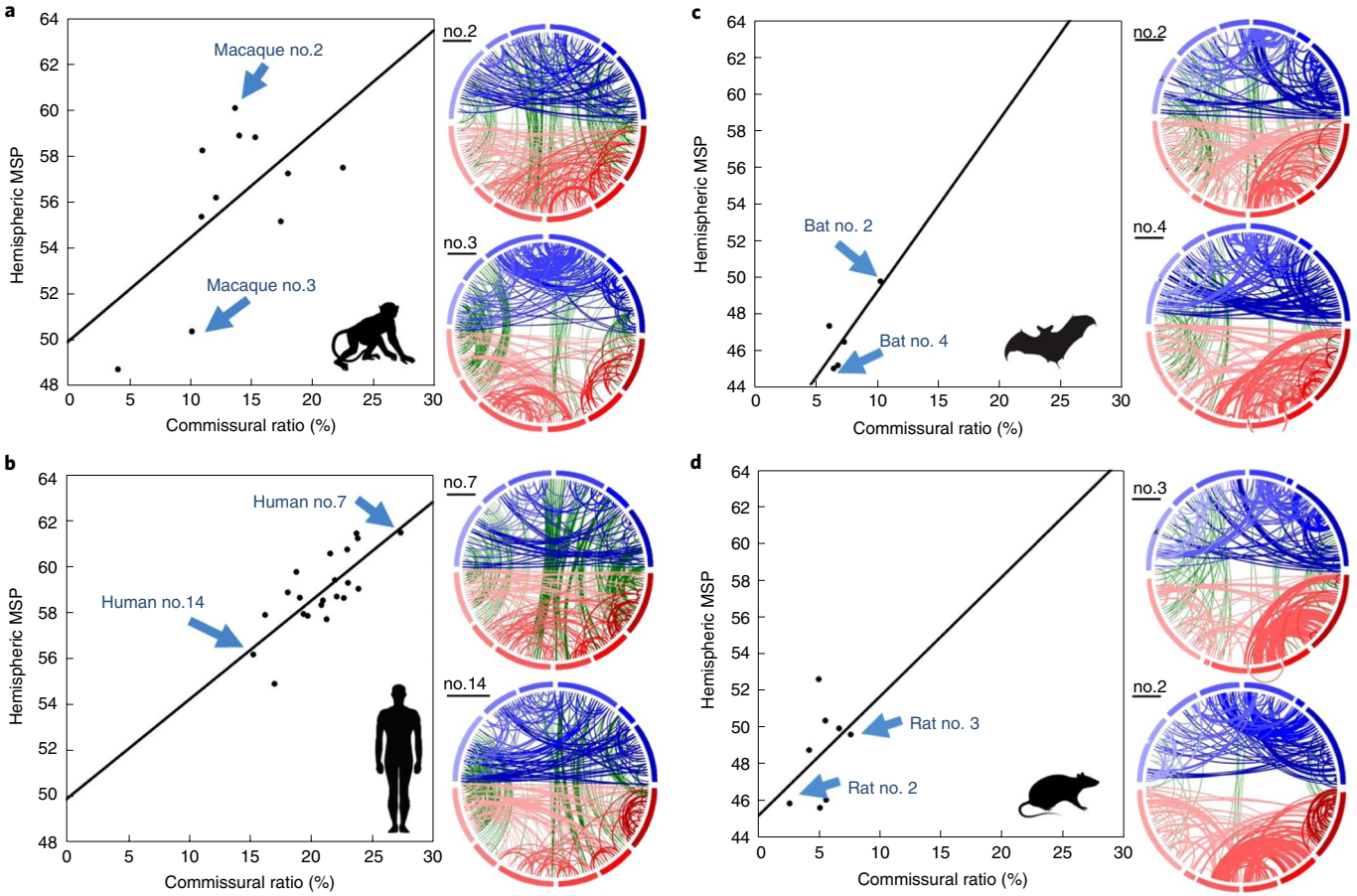


Fig. 2 | Intraspecific connectivity conservation. a–d, Intrahemispheric MSP as a function of commissural proportion in macaques (**a**), humans (**b**), bats (**c**) and rats (**d**). Two representative connectomes are presented in the form of a circular graph showing the connections of all 200 nodes. Each two connectomes show a low intrahemispheric MSP (bottom) and a high intrahemispheric MSP (top), with the left hemisphere connections in red and the right ones in blue. The interhemispheric connections are shown in green. The nodes were arranged according to their anatomic location in the brain (Extended Data Fig. 1). Comparing the circular graphs of high MSP individuals with those of low MSP individuals reveals that, in low MSP graphs, there are fewer green commissural connections and stronger blue and red intrahemispheric connections (depicted by thicker lines; Pearson's correlation effect sizes for macaques, humans, fruit bats and rats, respectively; $R^2 = 0.31$, $n = 11$; $R^2 = 0.57$, $n = 22$; $R^2 = 0.22$, $n = 5$; $R^2 = 0.44$, $n = 7$). All statistical analyses in this figure were two tailed.

longitudinal fasciculus is among the strongest connections in all species. We thus reconstructed it in 48 of the species and found a significant negative correlation between its weight and the MSP, suggesting that it contributes to improved connectivity (Fig. 3b; $R^2 = 0.39$, $P < 10^{-5}$, Pearson's correlation t-test, $n = 48$).

The forces and constraints that drove the evolution of the mammalian brain, including our own human brain, are still far from understood. In the present study, we reveal a conservation principle that allows network connectivity and normalized wiring cost to be fixed almost independent of network architecture. The comparison of the normalized (and not the absolute) wiring cost is reasonable because larger animals can invest more energy in wiring. This conservation principle holds for all mammals, independent of brain size, suggesting a selective evolutionary pressure in favor of efficient connectivity and wiring costs. This conclusion is further supported by our observation that the conservation principle applies not only to inter- but also to intraspecific variations of the connectome. DTI has several well-known limitations[17,18], but our many controls (see Methods) suggest that these limitations could not be the cause of the phenomenon that we describe in the present study.

Our results show that some species have more interhemispheric connections and fewer connected hemispheres, and vice versa. This same pattern also holds within species, including humans (the slope of the MSP to commissural proportion in humans was no different from that of all mammals; $P = 0.35$, Chow's test with 27 humans and 226 non-human specimens). It is not known what drives the brain of an individual or a species to be on one or other side of this spectrum and what the cognitive and clinical implications are. These are some of the intriguing questions that arise from our study. A recent study showed that human brains of patients with agenesis of the corpus callosum have stronger intrahemispheric connections, underscoring the same conservation principle[19], suggesting that it plays a role during the ontogeny of the connectome. Similar observations were found for individuals who underwent childhood hemispherectomy[20].

More than 100 years ago, the pioneer of brain connectivity research, Ramon y Cajal, hypothesized that brains should conserve wiring costs and conductivity speed[11]. Using an immense comparative approach, we were able to confirm these predictions. The database collected for the present study is expected to reveal many more fundamental principles of mammalian connectivity, some of which probably translate to cognitive and other functionalities. Deviations of species and individuals from these principles might imply clinical conditions or unique cognitive functionality. As mammals, humans should have a special interest in these principles.

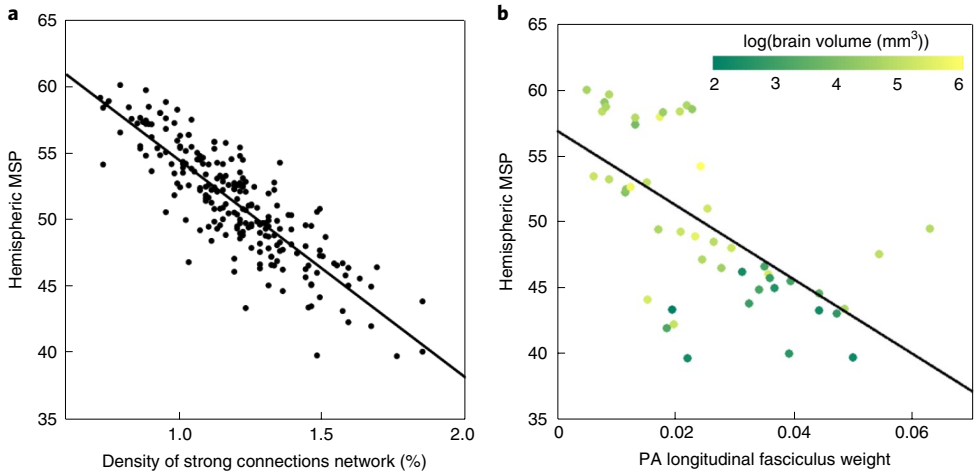


Fig. 3 | Strong connections drive hemispheric connectivity. a, Hemispheric MSP negatively correlates with the density of the strong-connections network ($P < 10^{-61}$, Pearson's correlation *t*-test, $R^2 = 0.71$, $n = 232$). To reveal the identity of the specific strong connections, we generated a heatmap showing all strong connections projected on to a midsagittal slice (Extended Data Fig. 2). The estimated landmarks of the four lobes are shown for comparison. This heatmap suggests that the posterior–anterior longitudinal fasciculi (light red colors in Extended Data Fig. 2) are the strongest connections across mammals, regardless of brain volume or MSP. **b**, Hemispheric MSP as a function of the weight of the posterior–anterior (PA) longitudinal fasciculus in 48 species (the volume of which is color coded ($P < 10^{-5}$; Pearson's correlation *t*-test, $R^2 = 0.49$, $n = 48$). All statistical analyses in this figure were two tailed.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41593-020-0641-7.

Received: 3 October 2019; Accepted: 10 April 2020;
Published online: 8 June 2020

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Methods

Brain samples. The mammalian MRI (MaMI) database includes diffusion and T2-/T1-weighted scans of 123 different species, with replicas of some of the species reaching a total of 232 brains (Supplementary Table 1). No statistical methods were used to predetermine sample size. We acquired as many samples as possible, far more than previous comparative approaches. All scans (except for the human participants) were performed on excised and fixated tissue. Scans were performed using either a 7-T 30/70 Biospec Avance Bruker system or a 3-T Siemens Prisma system. All brains that fitted the bore of the 7-T system were scanned in it, whereas the larger brains were scanned in the 3-T scanner. No animals were deliberately euthanized for the present study. Brain collection was based on incidental death of animals in zoos in Israel and naturally dying animals collected abroad. All animals were collected with the permission of the national park authority (approval no. 2012/38645) or its equivalent in the relevant countries. Animals' brains were extracted within 24 h of death and placed in formaldehyde (10%) for a fixation period of a few days to a few weeks (depending on the brain size). Some 24 h before MRI, the brains were placed in phosphate-buffered saline for rehydration. For the MRI, the brains were placed in a plastic bag and immersed in fluorinated oil (Flourinert, 3 M). This procedure is done to minimize image artifacts caused by magnet susceptibility effects.

Human data. The MaMI database includes five humans (three males and two females) and were used for all the mammalian comparative analyses (Indicated on Fig. 2a). For the within-species comparisons (Fig. 3), we used an additional database of 22 humans (11 females and 11 males, with a mean age of 37 ± 9 years). No statistical methods were used to predetermine the sample size, but our sample size is similar to that reported in previous publications[21]. All the humans ($n = 27$ in total) were neurologically healthy and signed an informed consent approved by the institutional review board of Sheba Medical Center and the ethics committee of Tel Aviv University. The analysis pipeline was similar to the one used in the MaMI database.

MRI acquisition. The MRI protocol was similar for all species and included high-resolution anatomical scans (T2- or T1-weighted MRI), which were used as an anatomical reference, and diffusion MR scans. The diffusion MRI included acquisition by high angular resolution diffusion imaging (HARDI). The HARDI dataset consisted of a series of diffusion-weighted, spin-echo, echo-planar-imaging images covering the whole brain, scanned in either 60 (in the 7-T scanner) or 64 gradient directions (in the 3-T scanner) with an additional 3 non-diffusion-weighted images (B0). The b value was 1,000 s mm^{-2} in all scans. In the 7-T scans, TR was longer than 12,000 ms (depending on the number of slices), TE was 20 ms and $\Delta/\delta = 10/4.5$ ms. In the 3-T scans, TR was 3,500 ms, with a TE of 47 ms and $\Delta/\delta = 17/23$ ms. We kept the matrix size constant across all species (128×96) so that the two-dimensional image pixel resolution (per slice) was linearly scaled according to brain size. We ran a control analysis to make sure that this decision did not influence our results (see "DTI analysis and tractography"). The number of slices varied due to differences in brain shape and was between 46 and 68. The number of scan repetitions and the acquisition time were different for each species and depended on the brain size and the SNR. We tried to keep the SNR >20, resulting in acquisition times of 48 h for the small brains (~0.15 ml) and 25 min for the large brains (>1,000 ml). SNR was defined as the mean signal strength of the brain divided by the s.d. of the noise (an area in the non-brain part of the image).

DTI analysis and tractography. For HARDI/DTI analysis and tractography we used ExploreDTI[22] software. The following steps were used to generate the fiber tracts:

(1) Anisotropic smoothing (using a 3-pixel Gaussian kernel) was carried out to reduce noise and smooth the data.
(2) Motion and distortion correction: motion, susceptibility and eddy current distortions were corrected in the native space of the HARDI acquisition.
(3) Fiber-orientation density functions were generated, per pixel using the spherical harmonic deconvolution approach[23], providing possible multiple ($n \geq 1$) fiber orientations per voxel. We used spherical harmonics up to the fourth order. We verified that the result also held when using traditional DTI-based tractography (Supplementary Fig. 8).
(4) Whole-brain tractography was performed using a CSD seed point threshold similar for all samples (0.2) and a step length half the pixel size.

The output of this step is a list of streamlines (estimated fiber tracts) starting and ending at any of the voxels (including their full trajectory). All brains that could be scanned were used in the following analyses with no exclusions.

Network generation and analysis. All analyses were performed using Matlab 2018 (Mathworks). As we were interested in inner-cerebrum connectivity, external projection fibers (that is, passing through the cerebral peduncle) as well as cerebellar connections were manually removed for each brain. Other inner-hemispheric projection fibers (such as the thalamic radiation) were included in the analysis. The midline of the brains was manually marked, and the tracts were divided into three groups: left hemisphere, right hemisphere and interhemispheric tracts. To parcellate the brain into 200 nodes, all of the fiber end-point (x–y–z) positions were used as input for a k-means algorithm. The advantage of the k-means algorithm over a random node division is that vertices with a similar connectivity profile would group together. The k-means algorithm also helped when dealing with unconnected network components (Supplementary Fig. 9). This was performed twice—once for each hemisphere. The center of each of the 200 clusters was taken as a node in the network. Previous studies have shown that fiber tracking can cause bias where the vast majority of fiber end-points reside in the white matter[24]. Our tracking approach (which included CSD tracking of the HARDI dataset) assured that the vast majority of tracking end-points (~90%) reside in the cortical or subcortical gray matter (see examples in Supplementary Fig. 12). To compute the edges of the network, we randomly selected 4,000 fibers per hemisphere (from the full list of fiber tracts). We used a fixed number of fibers per hemisphere across all species, to avoid biases resulting from differences in the number of fibers (due to differences in scan quality). Next, a connectivity matrix was generated indexing the number of streamlines (tracts) between any two nodes[25]. We repeated this procedure 100 times, each time randomly selecting 4,000 fibers per hemisphere (we validated that changing this number did not affect the results; Supplementary Fig. 8). All of the network analyses (see "Controlling for the scanning resolution and acquisition parameters") were thus performed 100 times and their average was taken. The connectivity matrix of each hemisphere was used to compute the intrahemispheric MSP (the left and right hemispheric networks were estimated separately and then averaged). As we used the end-points of the fibers as nodes, the largest connected component of the network was the entire network. In addition, we computed the mean distance (tract length) between every node-pair as the mean distance of all fibers connecting the two. This fiber length was also used to compute the wiring cost index per brain (the sum of the multiplication of the connectivity matrix and the length matrix for 4,000 connections sampled across the entire brain). Note that we sampled an equal number of fibers from all species, although in reality this number is probably not conserved across species. Our cost measurement can thus be thought of as the cost for a constant number of fibers. Data analysis and collection were not performed blind because we did not divide the data into groups.

Controlling for the scanning resolution and acquisition parameters. As we kept the matrix size constant across all species, voxel dimensions were linearly scaled with brain volume. This resulted in different scanning resolutions across the samples. We performed several analyses to verify that this was a reasonable decision including: (1) comparing our DTI-based connectome with the connectome of the house mouse (*Mus musculus*) generated using track tracing (see "Reconstruction of the track-tracing mouse connectome"). The strong correlation between the two networks suggests that our method can reconstruct the essence of the connectome (Supplementary Fig. 13). (2) For nine species, we scanned the brains in multiple resolutions so that we could perform the analyses with a comparative absolute resolution (see "Controlling for the scanning resolution" and Supplementary Figs. 5 and 6). (3) We show that we can reconstruct specific fiber systems across brains and that the bundles scale in size with the brain volume (Supplementary Figs. 14 and 15). (4) We analyze the effect of fiber crossing in the brain on our results (Supplementary Fig. 16). (5) Finally, for four species for which we had more than four specimens, we showed that the connectivity conservation principle is also observed within species, where absolute resolution was almost identical (Fig. 3).

Reconstruction of the track-tracing mouse connectome. We used the Allen mouse connectivity data[26] to create a tract-tracing connectome that can be compared with our tractography-based connectomes. To align the two connectomes on the same atlas coordinate system, we used a mouse brain atlas that was segmented into 20 cortical and subcortical regions (Supplementary Fig. 13a). Each of the injection and projection sites of the Allen mouse connectivity atlas was identified and assigned to each of these 20 regions. The weight of the connection between each pair of regions was defined as the sum of the connectivity strength according to the Allen mouse connectivity data. For the tractography-based mouse connectome, we used the mouse data that were part of our database following registration to the same atlas. The connectivity weights in this connectome were defined as the number of streamlines passing between each pair of regions. The two connectomes are shown in Supplementary Fig. 13b,c. Mantel's test for correlation between matrices was highly significant ($P = 2.3 \times 10^{-4}$).

Controlling for the scanning resolution. The brains of nine species (human, lama, macaque, green monkey, hare, wild rat, common fox, howler monkey and baboon) were scanned at higher resolution compared with the one used for the entire database (for example, 1.3 mm^3 for humans with all others being scaled accordingly). Each scan was then down-sampled three times (by factors of 1.5, 2 and 3). The above-mentioned network analysis was repeated for these nine brains at all four resolutions. The relationship between the hemispheric MSP and the commissural ratio was maintained across the different resolutions (Supplementary Fig. 5). To further investigate those data, we performed a linear regression analysis between each animal hemispheric MSP and data resolution. A linear relationship was found in all animals (Supplementary Fig. 6a), suggesting that we can estimate the hemispheric MSP for any resolution. We have created an estimated hemispheric

MSP for each of the nine animals as if it had been scanned at a resolution of 0.15, 1 or 2 mm (Supplementary Fig. 6b). The relationship between the hemispheric MSP and the commissural ratio remained highly significant. Finally, we acknowledge the fact that the difference in scanning resolution between the largest brain (human, 1.3 mm) and that of the wild rat (0.15 mm) was almost a factor of 10. To reduce this factor, we took the closest match to the human resolution for the multi-resolution dataset (Supplementary Fig. 6c). In that analysis the difference in data resolution reduced to a factor of 2–3, yet the linear relationship between the hemispheric MSP and the commissural ratio remained highly significant.

Reconstructing fiber systems and bundles. DTI-based fiber-tracking procedures measure large-scale connections. It allows the reconstruction of fiber fascicles: white matter structures usually with a cylindrical symmetry of a few tens to hundreds of micrometers in diameter. Such meso-scale structures are usually larger than the acquired resolution, thus allowing reconstruction of fascicle systems across species (as could also be learned from the comparison with the track-tracing connectome; Supplementary Fig. 13). To demonstrate this ability, Supplementary Figs. 14 and 15 show the reconstruction of the posterior–anterior longitudinal fasciculus, corpus callosum and anterior thalamic radiation in nine different species, spanning the entire range of brain volumes and thus of acquired image resolutions in the present study.

We performed another analysis to show this scaling effect: in the hierarchy of white matter organization, axons in the brain are grouped into bundles, which are then further grouped into fascicles. Mostly, individual bundles within fascicles cannot be segmented using MRI. However, in one specific case, bundles that are part of the corona radiata, passing within the lentiform (putamen), can be visualized using high-resolution, anatomical MRI. We thus measured the diameter of these bundles across four species with substantially different brain volumes and found that the bundle size scales with the brain volume (Supplementary Fig. 16). This observation implies that the size of the brain determines the cross-sectional size of central nervous system axonal bundles, supporting our relative equal resolution scanning.

Multiple fiber-orientation analysis. To reveal multiple fiber-orientations (MFOs), we used a constrained, spherical, harmonics deconvolution approach for fiber tracking. Using this analysis, we could compute the fraction of voxels with MFOs relative to the brain volume (Supplementary Fig. 17a). We found a linear relationship between the fraction of MFOs and the brain volume, suggesting that larger brains have more crossing fibers. However, the two fiber-tracking methods that we used (spherical harmonics and conventional DTI) both revealed the same conservation principle (compare Supplementary Fig. 8a and Fig. 2), even though the first can reveal crossing fibers whereas the second cannot. This suggests that the issue of crossing fibers cannot explain our main observation.

Furthermore, an analysis of one of the hallmark regions of crossing fibers in the human brain (the intersection between the forceps minor fibers and other frontal projection/association fibers) revealed that the crossing fibers in this region can be identified in all animals that were tested regardless of their brain volume (Supplementary Fig. 17b,c).

Estimating the proportion of commissural fibers. For each brain, the commissural fibers were identified (only the main three systems: the corpus callosum, anterior commissure and hippocampal commissure, which account for the vast majority of commissural connections in all mammals). This was done by placing a seed tractography region of interest in the midsagittal plane. The number of fibers found in this way was divided by the total number of fibers, to estimate the proportion of commissural fibers (commissural proportion). As this proportion might be biased by tracking parameters, we computed an additional estimate of the commissural ratio[9] by dividing the (square root of the) midsagittal area of commissural systems (manually segmented) by the (cubic root of the) brain volume,

Gray and white matter volume calculation. As we scanned brains without skulls, the total brain volume was computed by summing the number of voxels above a noise threshold multiplied by the image resolution. To estimate gray and white matter volumes, we fed the fractional anisotropy maps to a Gaussian mixed-model analysis, which divided the voxels into three tissues (white matter, gray matter and cerebrospinal fluid). To estimate white and gray matter volume, the probabilities of the voxels (belonging to these two tissue types) were summed over the entire brain and multiplied by the image resolution.

An exception was the human brain scans (which were done in vivo), for which SPM12 was used to perform skull stripping and segmentation to create a brain mask.

Controlling for the number of nodes. In the present study, we segmented the brain into 100 nodes per hemisphere. To verify that the number of nodes had no effect on the results, we ran the network analysis described while parcellating the brains to a different number of nodes, from 50 to 400 in increments of 50.

Statistical analysis. All statistical analyses were performed using Matlab 2018 (Mathworks). We used Pearson's correlation coefficient to correlate various network (for example, MSP) and brain parameters (for example, brain volume) using $P < 0.05$ as the level of significance. All our statistics were two tailed. We did not test for the normality of the data because this is not required for a large sample size such as we had. As we did not correct for the phylogenetic dependency of the samples, we also ran Pearson's correlation for all mammalian orders. To control for replicates (multiple specimens of the same species) we used a GLM to test the relationship between the hemispheric MSP and the commissural proportion. The GLM was run with both the commissural proportion and the brain volume taken as fixed explaining factors and the MSP set as the explained variable. We used Chow's test to compare the coefficients of the MSP to commissural-proportion regressions in humans and all mammals. We did not use any randomization of the sample, because the data were not grouped into any 'conditions' apart from according to their phylogenetic orders. For more details on the analyses decisions see Nature Research Reporting Summary.

Figure axes. We present only the relevant range of the data in the figures, which means that, in some cases, axes do not start at '0'. Stretching the graphs to '0', far beyond the relevant range of the data, would make them hard to read and is unnecessary in our case because we are not comparing several groups or treatments but examining the slope.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding author upon request.

Code availability

All codes used in this study are part of Matlab's (Mathworks) statistics and data analysis toolbox codes. The circular graphs were generated using P. Kassebaum's (2020) circularGraph (https://www.github.com/paul-kassebaum-mathworks/circularGraph).

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Acknowledgements

We thank K. Or, R. Elazari, S. Landau, A. Scheinin, D. Morik, N. Avni-Magen, N. Kornfeld-Schor, I. Horowitz and A. Kafri for help with brain collection. We also thank A. Leemans for help with the fiber-tracking procedures. Y.A. thanks the Strauss Center for Computational Neuroimaging for help in data acqusition.

Author contributions

Y.A. was responsible for data acquisition and data analysis. O.Z. was responsible for code and software development. A.M. and A.B. were in charge of sample collection and preparation. Y.Y. was responsible for study and analysis design and wrote the paper, together with Y.A. All authors contributed to the discussion and interpretation of the results.

Competing interests

The authors declare no competing interests.

Additional information

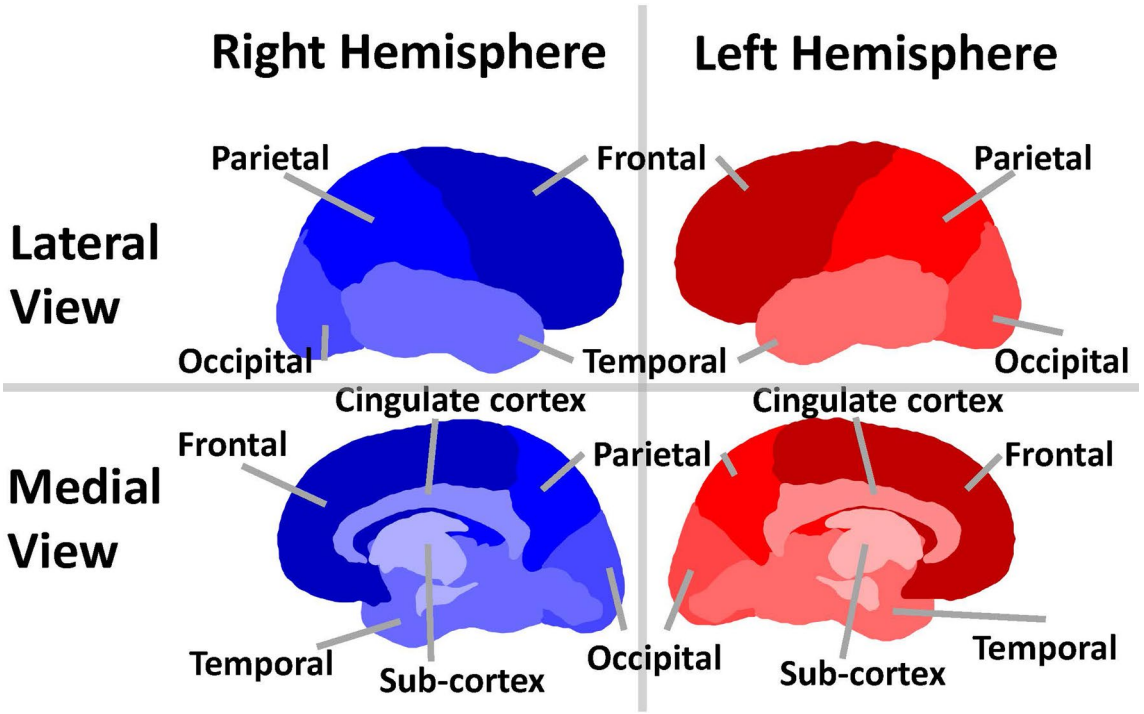
Extended data is available for this paper at https://doi.org/10.1038/s41593-020-0641-7.

Supplementary information is available for this paper at https://doi.org/10.1038/s41593-020-0641-7.

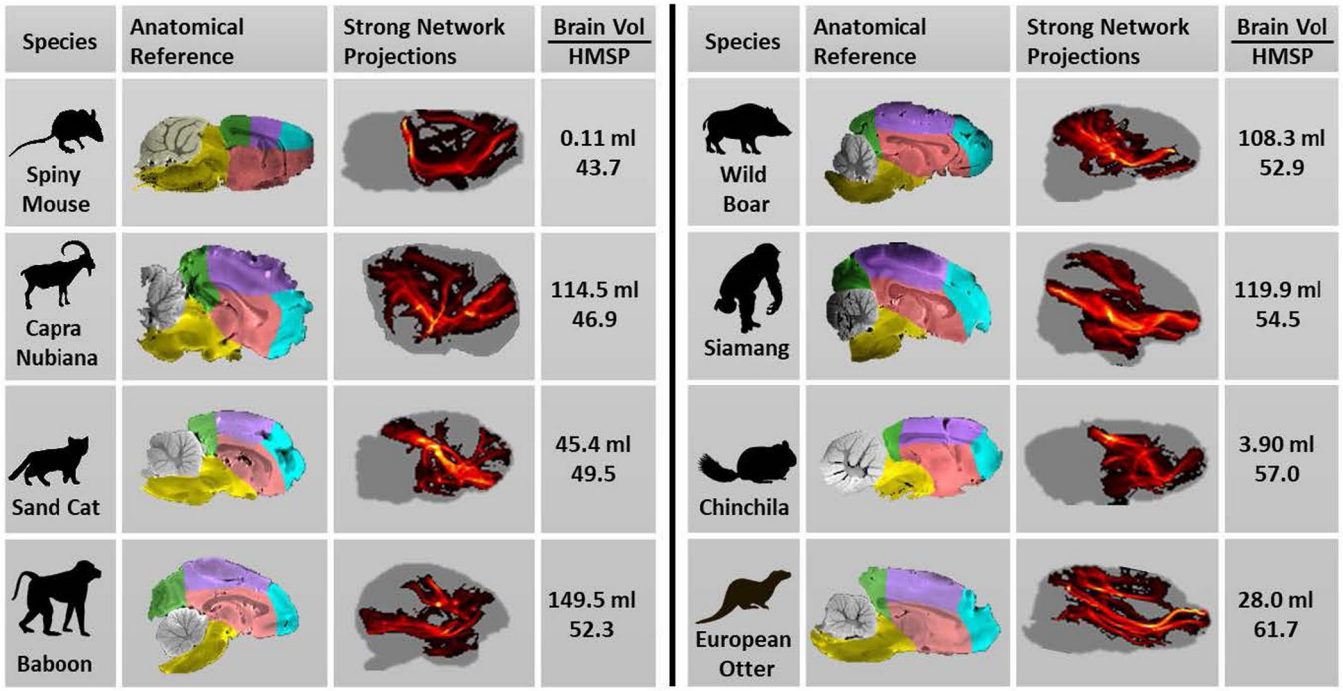
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Peer review information *Nature Neuroscience* thanks Daniel Alexander, Rogier Mars and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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Extended Data Fig. 1 | Brain Regions. Illustration of the different brain regions that correspond to the circular graphs shown in Fig. 1 matched by color and shading.



Extended Data Fig. 2 | Strong Connections. A heat-map showing all strong connections projected on-to a mid-sagittal slice.

nature research

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Last updated by author(s): Apr 9, 2020

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☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	MRI - either the 7T Biospec 30/70 Avance spectrometer or the 3T siemens Prisma Scanner
Data analysis	Matlab 2018, SPM12, Explorer DTI 4.8.6

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We aimed to collect data from as many species as possible. this is by far the largest sample size ever collected
Data exclusions	All brains that could be scanned were used in the study (with no exclusions)
Replication	when possible, we scanned multiple specimens per species and we used these replications to test the robustness of the results. All replication tests revealed the same results
Randomization	We did not use any randomization, as data was not grouped into any 'treatments' except for their phylogenetic orders.
Blinding	Data analysis and collection were not performed in a blind manner as we did not divide the data into groups.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	Rattus norvegicus - wistar and Mus musculus - icr. We used both males and females of ca. 1 year old
Wild animals	we have more than 100 species - all names are provided in the manuscript
Field-collected samples	Brains were collected from dead animals. No animals were deliberately euthanized for this study. Brain collection was based on incidental death of animals in Zoos in Israel and on naturally dying animals collected abroad. All animals were collected with permission of the national park authority (approval no. 2012/38645) or its equivalent in the relevant countries. Animals' brains were extracted within 24 hours following death and placed in Formaldehyde (10%) for a fixation period of a few days to a few weeks (depending on the brain size). Twenty-four hours prior to MRI scan, the brains were placed in phosphate buffered saline (PBS) for rehydration. As we only received the brains, after animals died, there was no need for housing, maintenance, etc.
Ethics oversight	We had permission from the National park authorities to collect brains and from the TAU IACUC to use laboratory animals (permit 04-18-020).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about studies involving human research participants

Population characteristics	13 males, 14 females, mean age 37 +/-9. Israelis
Recruitment	Through an advertisement according to the IRB approval
Ethics oversight	Sheba medical center IRB and TAU IRB

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Magnetic resonance imaging

Experimental design

Design type	Pre-clinical 7T (Biospec 70/30, Bruker, Germany) and clinical 3T (Prisma, Siemens, Germany)
Design specifications	The MRI protocol was similar for all species and included high resolution anatomical scans (T2 or T1 weighted MRI) that were used as anatomical reference and diffusion MRI scans. The diffusion MRI included high angular resolution diffusion imaging (HARDI) acquisition. The HARDI dataset consisted of a series of diffusion weighted spin-echo echo planar imaging images covering the whole brain scanned at 60 gradient directions (in the 7T scanner) and 64 gradient directions (in the 3T scanner) with additional 3 non-diffusion weighting images (B0). The b value was 1,000 s/mm2 in all scans. In the 7T scans TR was longer than 12000ms (depending on number of slices) and TE was 20ms, Big/Small delta were 10/4.5ms. In the 3T scans TR was 3500ms, with TE of 47ms, small/big delta were 17/23ms. We kept the matrix size constant across all species (128x96) so the 2D image pixel resolution (per slice) was linearly scaled according to brain size. We ran a control analysis to make sure that this decision did not influence our results (see below). The number of slices varied due to differences in brain shape and was between 46-68. The number of scan-repetitions and the acquisition time was different for each species and depended on brain size and on signal to noise ratio (SNR). We tried to keep SNR above a value of 20 resulting in acquisition times of 48 hours for the small brains (~0.15ml) and 25 minutes for the large brains (>1000ml). SNR was defined as the mean signal strength of the brain divided by the SD of the noise (an area in the non-brain part of the image).
Behavioral performance measures	There was no behavioral study

Acquisition

Imaging type(s)	structural and diffusion MRI
Field strength	3T or 7 T
Sequence & imaging parameters	The MRI protocol was similar for all species and included high resolution anatomical scans (T2 or T1 weighted MRI) that were used as anatomical reference and diffusion MRI scans. The diffusion MRI included high angular resolution diffusion imaging (HARDI) acquisition. The HARDI dataset consisted of a series of diffusion weighted spin-echo echo planar imaging images covering the whole brain scanned at 60 gradient directions (in the 7T scanner) and 64 gradient directions (in the 3T scanner) with additional 3 non-diffusion weighting images (B0). The b value was 1,000 s/mm2 in all scans. In the 7T scans TR was longer than 12000ms (depending on number of slices) and TE was 20ms, Big/Small delta were 10/4.5ms. In the 3T scans TR was 3500ms, with TE of 47ms, small/big delta were 17/23ms. We kept the matrix size constant across all species (128x96) so the 2D image pixel resolution (per slice) was linearly scaled according to brain size. We ran a control analysis to make sure that this decision did not influence our results (see below). The number of slices varied due to differences in brain shape and was between 46-68. The number of scan-repetitions and the acquisition time was different for each species and depended on brain size and on signal to noise ratio (SNR). We tried to keep SNR above a value of 20 resulting in acquisition times of 48 hours for the small brains (~0.15ml) and 25 minutes for the large brains (>1000ml). SNR was defined as the mean signal strength of the brain divided by the SD of the noise (an area in the non-brain part of the image).
Area of acquisition	Whole brain

Diffusion MRI ☒ Used ☐ Not used

Parameters: We have used: 60 gradient directions (in the 7T scanner) and 64 gradient directions (in the 3T scanner) with additional 3 non-diffusion weighting images (B0). The b value was 1,000 s/mm2 in all scans. No cardiac gating was applied

Preprocessing

Preprocessing software	Matlab 2018 + Explore DTI 4.8.6+ SPM12
Normalization	No normalization was applied
Normalization template	No normalization template was applied
Noise and artifact removal	Motion + Eddy current correction
Volume censoring	We did not use volume censoring

Statistical modeling & inference

Model type and settings	We used regression analysis (Pearson) and GLM
Effect(s) tested	We analyzed correlations between different graph parameters

Specify type of analysis: ☒ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference (See Eklund et al. 2016)	We analyzed correlations between different graph parameters
Correction	None - there were no multiple tests. We did not correct for phylogeny and refer to this in the methods

Models & analysis

n/a	Involved in the study
☒	☐ Functional and/or effective connectivity
☐	☒ Graph analysis
☒	☐ Multivariate modeling or predictive analysis

Graph analysis	we used weighted graphs. Species were the dependent variables. We used the following network parameters: mean short past and normalized cost (edge length).